

Weekly Report: [W08]

General Information

- **Group ID:** 05
- **Group Name:**
- **Class:** 25A01
- **Project Name:** Super Mario Game
- **Date range:** 2026-07-27 – 2026-08-01
- **Github Repository:** [TBD]

Tasks Completed This Week

25125032 - Lê Minh Nhật

- Implemented sprite sheet animation timer and frame selection logic for Mario's states (idle, running, jumping).
- Fixed visual artifacting and tile gap rendering issues in the SFML tilemap display layer.
- *Evidence:* Github commit history

25125046 - Vũ Mai Thùy

- Implemented sprite sheet animation timer and frame selection logic for Mario's states (idle, running, jumping).
- Fixed visual artifacting and tile gap rendering issues in the SFML tilemap display layer.
- *Evidence:* Github commit history

25125017 - Nguyễn Đăng Khang

- Tuned physics movement formulas for Mario (horizontal acceleration, friction, jump impulses, and terminal velocity).
- Implemented Axis-Aligned Bounding Box (AABB) collision detection and resolution against static ground tiles.
- *Evidence:* Github commit history

25125065 - Bùi Viết Thành

- Expanded the core Entity class hierarchy to support state transitions (e.g., alive, inactive, interactable).
- Designed base sub-classes for interactable environmental entities (Question Blocks, Brick Blocks).
- *Evidence:* Github commit history

25125014 - Nguyễn Đức Huy

- Managed sprint milestones, updated task boards on GitHub Projects, and tracked task deliverables.
- Conducted weekly code reviews on physics-entity integration and updated software architecture documentation.
- *Evidence:* Github commit history

AI Usage Declaration

- Used AI to assist in fine-tuning velocity delta updates and gravity formulas for responsive jump physics.

Tasks Planned for Next Week

- **Task 1:** Implement horizontal camera tracking to keep the viewport focused on Mario as the level scrolls.
- **Task 2:** Implement base enemy entity classes (Goomba) using the entity hierarchy and set up enemy sprite sheets.

Issues

- **Issue 1:** Mario slightly clips into ground tiles when landing at high falling speeds.
- **Resolution:** Adjusted swept AABB collision checks to clamp vertical position directly to tile tops upon ground detection.